

Twitter Data Ingestion Pipeline - README

This project demonstrates a Data Engineering pipeline for Twitter sentiment data ingestion and streaming. Due to Twitter/X API limitations and paid access restrictions, a publicly available Twitter sentiment dataset was used to simulate real-time tweet ingestion.

Project Components

Dataset: Twitter Sentiment Dataset

Format: CSV converted to JSON

Streaming Tool: Apache Kafka

Programming Language: Python

Producer Script: twitter_producer.py

Completed Tasks

- Collected tweet dataset
- Converted CSV data into structured JSON format
- Prepared streaming-ready tweet dataset
- Designed Kafka producer pipeline for simulated real-time streaming

Files Included

- twitter_training.csv
- tweets.json
- twitter_producer.py
- README_Data_Ingestion_Project.pdf

Kafka Setup Instructions

1. Start Apache ZooKeeper
2. Start Apache Kafka Server
3. Create Kafka Topic:
`kafka-topics.bat --create --topic twitter_topic --bootstrap-server localhost:9092`

Run Producer Script

Execute the following command:
`python twitter_producer.py`

Project Workflow

CSV Dataset → JSON Conversion → Kafka Producer → Real-Time Streaming Simulation

Conclusion

The data ingestion phase has been successfully completed by transforming tweet data into structured JSON format and preparing the dataset for Kafka-based streaming implementation.